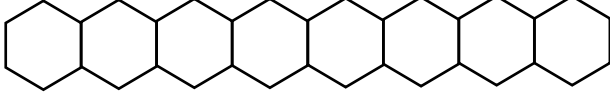


Series : ZYWX1

SET ~ 3

रोल नं.

Roll No.



कोड नं.
Q.P. Code

65/1/3

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

नोट / NOTE

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 3 हैं।
Please check that this question paper contains 3 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 12 प्रश्न हैं।
Please check that this question paper contains 12 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की एयर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख्य-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



MATHEMATICS

Integrals



निर्धारित समय: 1 घंटे

Time Allowed : 1 hour

अधिकतम अंक : 25

Maximum Marks : 25



General Instructions:

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **25** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **FOUR** sections- Sections **A, B, C** and **D**.
- (iii) In **Section A** : Question numbers **1** to **5** are Very Short Answer type questions. Each question carries **1** mark.
- (iv) In **Section B**: Question numbers **6** to **8** are Very Short Answer (VSA) type questions. Each question carries **2** marks.
- (v) In **Section C**: Question numbers **9** to **11** are Short Answer (SA) type questions. Each question carries **3** marks.
- (vi) In **Section D** : Question number **12** is a Long Answer question carrying **5** marks.
- (vii) There is no overall choice given in the question paper.
- (viii) Kindly note that there is a separate question paper for Visually Impaired candidates.
- (ix) Use of calculators is **NOT** allowed.

SECTION - A

This section contains 5 Very Short Answer Questions carrying 1 mark each.

1. Find the value of $\int_0^1 \frac{dx}{e^x + e^{-x}}$. 1
2. If $\int e^{-3\log x} dx = f(x) + C$, then find $f(x)$. 1
3. Find $\int 2^x + 2 dx$. 1
4. Find $\int \frac{2\cos 2x - 1}{1 + 2\sin x} dx$. 1
5. Find $\int \frac{\sin 2x \cos 2x dx}{\sqrt{9 - \cos^4 2x}}$ 1

SECTION - B

This section contains 3 Very Short Answer (VSA) type questions carrying 2 marks each.

6. Find: $\int \frac{\log x - 3}{(\log x)^4} dx$ 2



-
7. Evaluate : $\int_1^3 \frac{\sqrt{4-x}}{\sqrt{x} + \sqrt{4-x}} dx$ 2
8. Find: $\int \frac{\cos x}{\sin 3x} dx$ 2

SECTION - C

This section contains 3 Short Answer (SA) type questions carrying 3 marks each.

9. Find: $\int \frac{x - \sin x}{1 - \cos x} dx$ 3
10. Find: $\int \frac{\sqrt{x}}{1 + \sqrt{x^{3/2}}} dx$ 3
11. Evaluate: $\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)} dx$ 3

SECTION - E

This section consists of 1 Long Answer (LA) type question carrying 5 marks.

12. Evaluate: $\int_0^{3/2} |x \cos \pi x| dx$ 5

